

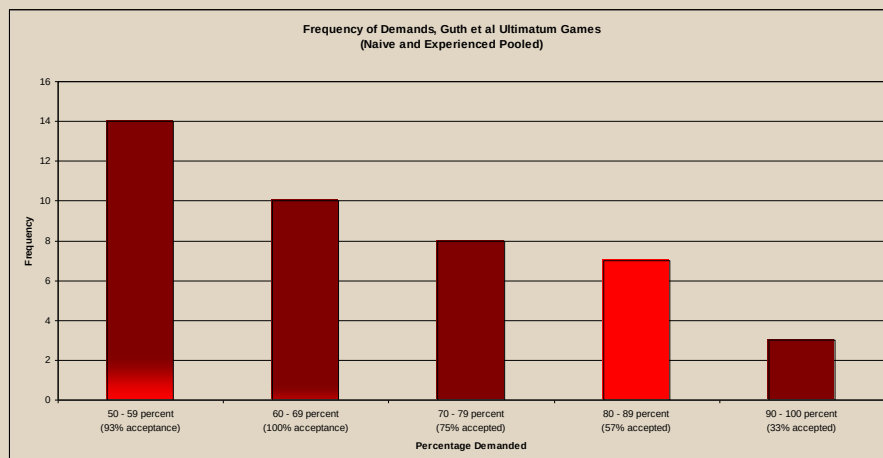
Social Preferences

John Van Huyck

Introduction

- Bargaining models often rely on strong assumptions
 - Common knowledge of rationality
 - Maximize their own monetary payoff
- Guth, Schmittberger, and Schwarze (1982) report an experiment testing these assumptions using the ultimatum game
 - A simple game
 - If the theory should ever work, it should work for ultimatum games

Guth et al. (1982) results



Introduction

- Basic results reject the joint hypothesis
- Recall the subgame perfect equilibrium prediction with own money motivated subjects
 - Responders have a strict incentive to accept any positive offer
 - Proposers should anticipate this and offer as little money as possible
- Spawned a huge literature
 - Proposers offer an average of 40 percent of the money
 - Responders reject small offers of 20 percent or so half the time.

Reaction: Doubt

- Methodological variables
 - Repetition
 - Repetition makes little difference
 - Slight tendency for offers and rejections to fall over time
 - Camerer wonders if this is learning or anger satiation
 - Stakes
 - Common criticism when theory doesn't do well
 - Generally, paying extra doesn't influence how hard people think when task is simple, like Ultimatum game
 - Several months wages have modest effect on rejections

Reaction: Acceptance

- Are proposers (player 1s) trying to be fair or are they reacting to strategic concerns, that is, is it a Nash equilibrium?
 - Forsythe et al (1994) dictator games
 - Culture studies

**Forsythe, Horowitz, Savin, and Sefton
(1994)**



- Dictator game – force the responder to accept whatever is offered.
- If proposers are making positive offers for strategic reasons, then offers should go to zero in the dictator game.
- Offers are substantially reduced, but over half of the proposers still make positive offers
- Dictator game results suggest that
 - strategic concerns account for some of the positive offers in ultimatum games,
 - But at least some subjects are motivated by concerns of fairness

Roth, Prasnikar, Okuno-Fujiwara, and Zamir (1991)



- Are there cultural differences in rejection rates and, hence, in offers in the ultimatum game?
- Looked at four countries,
 - United States (Pittsburgh),
 - Yugoslavia (Ljubljana),
 - Japan (Tokyo),
 - Israel (Jerusalem)
- Concerns
 - Compare behavior in markets and bargaining settings
 - Look for subject-pool differences
 - Use such differences to test and refine hypotheses about out-of-equilibrium behavior in ultimatum games

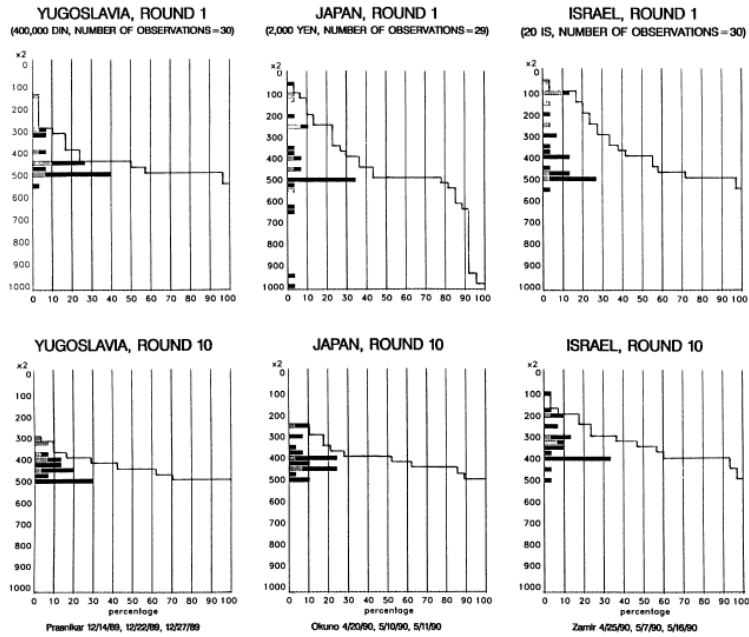


FIGURE 4. DISTRIBUTIONS OF BARGAINING OFFERS IN YUGOSLAVIA, JAPAN, AND ISRAEL (SOLID BARS = ACCEPTED OFFERS; STRIPED BARS = REJECTED OFFERS)

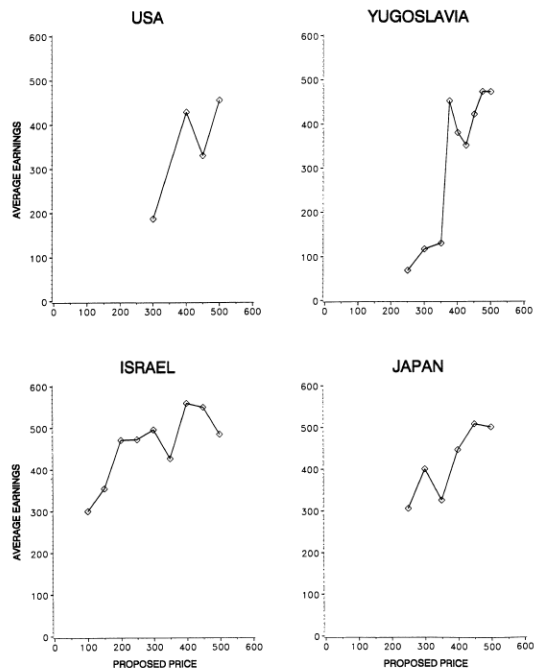


FIGURE 6. BUYERS' EARNINGS IN BARGAINING SESSIONS, BY PROPOSED PRICES (POOLED DATA OF 10 ROUNDS)

Roth et al. (1991)

- The modal proposal is
 - 500 in the United States and Yugoslavia
 - 400 in Japan and Israel
- Well calibrated with observed rejection behavior of responders (“sellers”)
 - Responders are tougher in the US and Yugoslavia than in Japan and Israel
 - Offers are lower in Japan and Israel because the profit maximizing offers is lower in these countries
 - The disagreement rates are similar across all four countries.

Roth et al. (1991) Market treatment

- 9 proposers (buyers) and 1 responder (seller)
- Proposers are attempting to buy an item worth 1000 tokens to them and nothing to the responder.
- Proposers simultaneously submit offers to buy to the responder in 5 token intervals (finite grid).
- The responder either accepts the highest offer or rejects them all.
- If he accepts it, the item is sold at the proposed price and all other proposers get zero.
- If he rejects it, everyone earns zero.
- The SPE has at least two proposers either proposing 995 or 1000 and the offer is accepted.

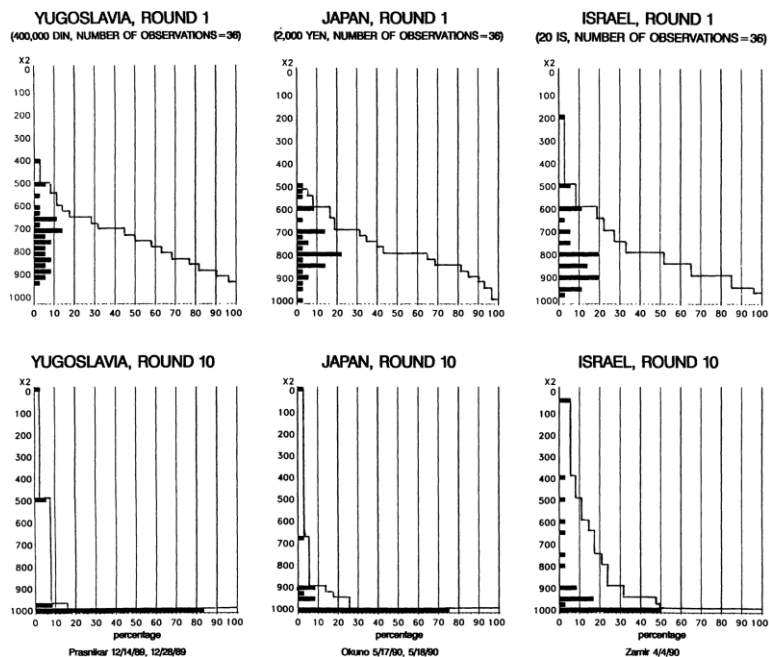


FIGURE 2. DISTRIBUTIONS OF MARKET OFFERS IN YUGOSLAVIA, JAPAN, AND ISRAEL

Roth et al. (1991) Market Treatment

- The SPE(s) makes an accurate prediction: convergence occurred by the 7th period.
 - All four countries result in very asymmetric earnings.
 - The responder (seller) gets all of the surplus.
- Contrast with the Ultimatum game:
 - Proposer competition eliminates equity concerns observed in the Ultimatum game.
 - Experience in the Ultimatum game eliminates low offers so behavior moves away from the SPE prediction.

Henrich (2002)

Table 2.5. Summary of ultimatum bargaining games

Group	Country	N	Stake size	Mean	Mode (percent of sample)	Standard deviation	Rejection rate (percent)	Rejection rate of <20 percent
Machiguenga	Peru	21	2.3	0.26	0.15/0.25 (72 percent)	0.14	4.8	10 (1/10)
Hadza (small)	Tanzania	29	1.0	0.27	0.20 (38 percent)	0.15	28	31 (5/16)
Tsimané	Bolivia	70	1.2	0.37	0.50/0.30/0.25 (65 percent)	0.19	0	0/5
Quichua	Ecuador	13	1.0	0.27	0.25 (47 percent)	0.16	015	50 (1/2)
Torguud	Mongolia	10	8.0	0.35	0.25 (30 percent)	0.09	5	0/1
Khazaks	Mongolia	10	8.0	0.36	0.25	0.09		
Mapuche	Chile	30	1.0	0.34	0.50/0.33 (46 percent)	0.18	67	20 (2/10)
Au	Papua New Guinea	30	1.4	0.43	0.30 (33 percent)	0.14	27	1/1
Gnau	Papua New Guinea	25	1.4	0.38	0.40 (32 percent)	0.19	40	50 (3/6)
Hadza (big)	Tanzania	26	1.0	0.40	0.50 (28 percent)	0.17	19	80 (4/5)
Sangu (farm)	Tanzania	20	1.0	0.41	0.50 (35 percent)	0.12	25	100 (1/1)
Unresettled	Zimbabwe	31	1.0	0.41	0.50 (56 percent)	0.14	10	33 (2/5)
Achuar	Ecuador	16	1.0	0.42	0.50 (36 percent)	0.20	0	0/1
Sangu (herd)	Tanzania	20	1.0	0.42	0.50 (40 percent)	0.09	5	1/1
Orma	Kenya	56	1.0	0.44	0.50 (54 percent)	0.092	4	0/0
Pittsburgh	USA	27	0.28	0.45	0.50 (52 percent)	0.096	22	0/1
Resettled	Zimbabwe	86	1.0	0.45	0.50 (70 percent)	0.10	7	57 (4/7)
Los Angeles	USA	15	2.3	0.48	0.50 (93 percent)	0.065	0	0/0
Ache	Paraguay	51	1.0	0.51	0.50/0.40 (75 percent)	0.15	0	0/8
Lamelara	Indonesia	19	10.0	0.58	0.50 (63 percent)	0.14	20	37

Henrich (2002)

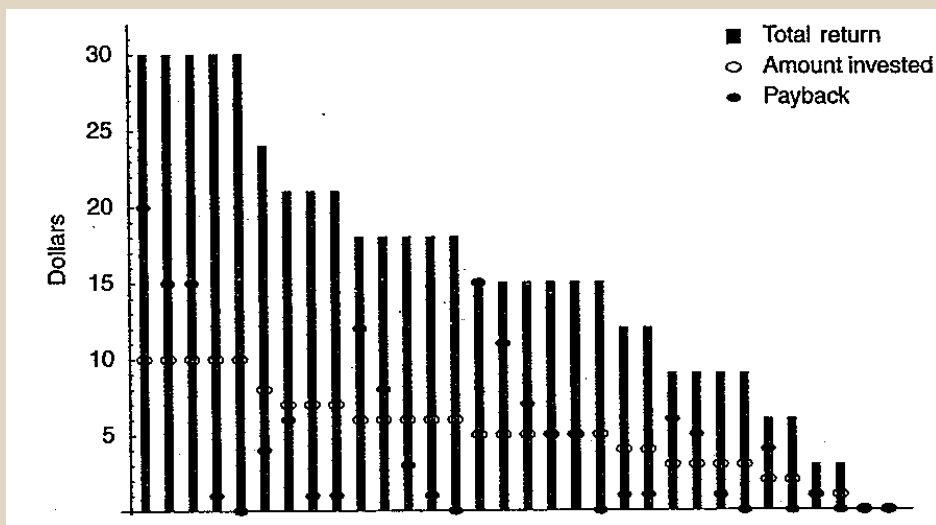
- Find two cultural variables that are correlated with differences in offers:
 - The amount of cooperative activity or economies of scale in production (hunting big game or whales)
 - The degree of market integration (an index combining the existence of a national language, a labor market for cash wages, and crop farming for cash).
 - Cultures with more cooperative activity and market integration have sharing norms closer to equal splits.
 - Suggests that “either market experience creates norms of equal division or the propensity to share evenly permits impersonal markets to flourish.”
- Camerer (2003, p. 74)

Berg, Dickhaut, and McCabe (1995)

- Investor endowed with X ,
 - which she invest, T , or can keep, $X - T$.
 - Investment earns a return $(1+r)$ and becomes $(1+r)T$
 - Trustee (dictator) divides the investment return $(1+r)T$ between himself and the investor.
 - Let Y denote how much he keeps and $(1+r)T - Y$ denote how much he returns.
 - The total payoff to the trustee is Y
 - The total payoff to the investor is $(X - T) + (1+r)T - Y$
- The SPE under own money motivated behavior is
 - Trustee keeps everything invested: $Y = (1+r)T$
 - Investor anticipating this invests nothing: $T = 0$

Berg, et al. (GEB, 1995): Trust Game

- $X = \$10$
- $r = 2$
- Elaborate mailbox design to ensure “double-blindness.”



Berg, et al. (1995): Trust Game

- Only 2 of 32 invest nothing as predicted by SPE.
- On average, Investors put in about 50 percent.
- The average amount repaid was about 95 percent of what was invested.
- The fact that the return to trust is around zero is fairly robust.

Van Huyck, Battalio, & Walters (GEB, 1995): “Peasant-Dictator Game”



- Same issue of GEB as Berg et al. (GEB, 1995)
- In fact, it is the very next article in the issue.
- Berg, et al. citations: 2,393
- Van Huyck, et al. citations: 66
- Camerer (2003, p.89): “These results [VHBW] are important because they find the *least* trust seen so far ...”

Van Huyck, et al. (GEB, 1995)

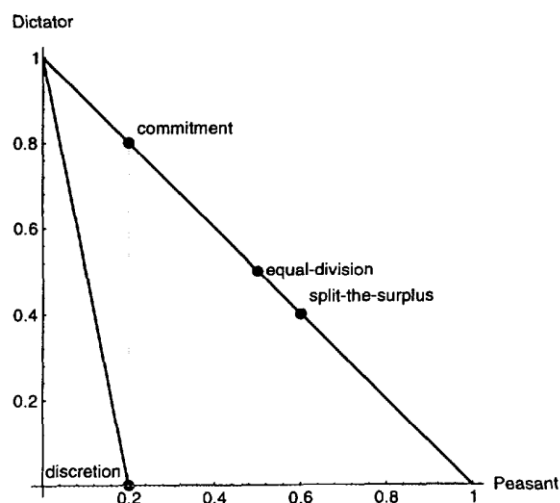


FIG. 1. Solution concepts illustrated in the payoff space; $W = 0.2$ and $r = 4.0$.

Van Huyck, et al. (GEB, 1995)

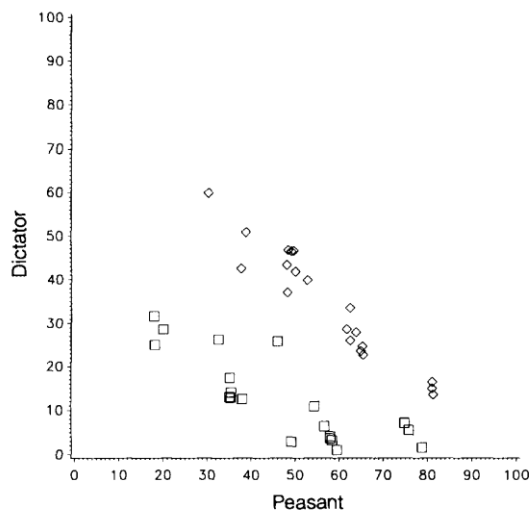


FIG. 2. Mean earnings plotted in the payoff space (denoted in cents). Square denotes discretion and diamond denotes commitment.

Van Huyck, et al. (GEB, 1995, p.151)

- “Figure 2 reports a scatter plot of the mean earnings in the last two treatments of each session.”
- That is, we threw away the first twenty periods of data!
- We did it because we interpret a Nash equilibrium as arising from enough experience for behavior (strategies and beliefs) to become coordinated on a fixed point of the best response correspondence.
- Behavior in early rounds appeared unstable to us.
- We focused on theory testing rather than measuring trust.

Camerer, (2003, p. 101)

- “Theorists worry-justifiably-that altering the utility function allows one to explain anything. One unpublished working paper concluded, ‘This [finding] puts the basis of our modeling on unobservable preferences, and raises the specter of extensive ad hoc modeling with a basis primarily in psycho babble.’”

Kinds of other regarding behavior

- Dictator game – presents evidence for altruism
- Ultimatum game – presents evidence for “anger,” “spite,” “negative reciprocity.”
- Market game – proposer competition eliminates the inefficiency from emotional reactions to unequal offers.
- Trust game – some evidence for “positive reciprocity.” (Dictator role returns about 1/3, which is more than straight dictator games.)

Social Preferences Theories

- Inequality-aversion
 - Fehr and Schmidt (1999)
 - Bolton and Ockenfels (2000)
- Fairness equilibrium
 - Rabin (1993)
 - Dufwenberg and Kirchsteiger (1998)
 - Falk and Fischbacher (2006)

Bolton and Ockenfels (2000) Example

- Consider an N person game. Let π_i be the monetary payoff for player i . Let Π be the vector of all players' monetary payoffs. The following function gives player i 's relative payoff. (Note that this is assuming non-negative payoffs.)

$$\sigma_i = \sigma_i(\Pi, N) = \frac{\pi_i}{\sum_{j=1}^N \pi_j} \quad \text{if } \sum_{j=1}^N \pi_j > 0$$

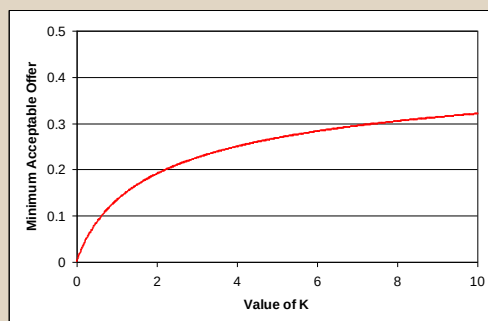
$$= \frac{1}{N} \quad \text{otherwise}$$

Bolton and Ockenfels (2000) ERC

- We can now define player i 's utility function as $u_i(\pi_i, \sigma_i)$.
 - By assumption, the first partial derivative is positive.
 - By assumption, holding π_i fixed, the function is maximized when $\sigma_i = 1/N$.
- For example

$$u_i(\pi_i, \sigma_i) = \pi_i - K \left(\frac{1}{N} - \sigma_i \right)^2$$

ERC and the ultimatum game



- As the split becomes very unequal, the value of the monetary payoff becomes small relative to the psychological loss of having to accept a small relative payoff. The responder is better off rejecting and getting the same payoff as the proposer instead of accepting and getting a higher monetary payoff. The minimum acceptable offer increases the more the responder cares about fairness.

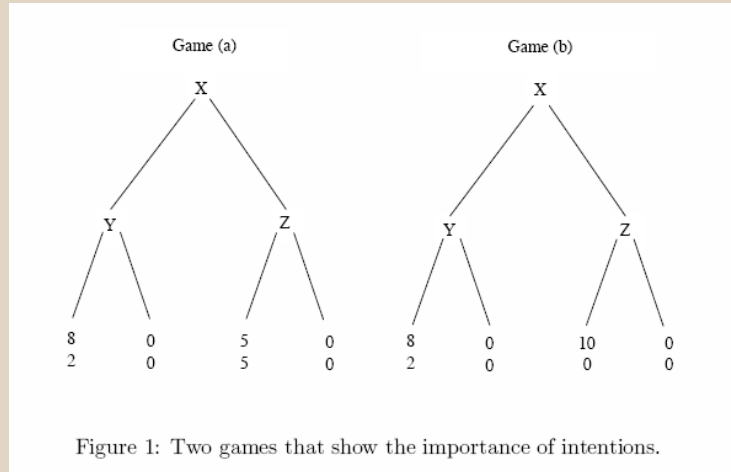
ERC and the market game

- In the market game, ERC predicts that all buyers will make close to the maximum offer.
- Suppose that highest offer being made is $M < 999$ tokens.
 - There must be at least one buyer who can gain by offering $M + 1$ tokens.
 - This is easiest to see for a buyer who is offering less than M tokens. This buyer would make a monetary payoff of zero. By offering $M + 1$ tokens, this buyer instead makes $1000 - (M + 1)$ tokens. His relative payoff goes up from zero to $1 - (M + 1)/1000$. On both payoff and relative payoff, this buyer is better off.
- Even if I'm doing badly versus the seller, making a high offer makes me better off versus the other buyers. This can make it worthwhile to make a high offer.

Against Consequentialism

- Most game theoretic models are consequentialist: preferences are defined over the consequences of the strategy combinations.
- The inequality aversion approach preserves consequentialism.
- Some evidence that intentions matter.

Falk, Fehr, and Fischbacher (1999)



Falk, Fehr, and Fischbacher (1999)

Table 2.6. *Ultimatum games with varying unchosen paths*

Unchosen offer	Interpretation of (8,2) offer	How often the (8,2) offer is	
		Rejected	Proposed
(5,5)	Relatively unfair	0.44	0.31
(2,8)	Not sacrificial	0.27	0.73
(8,2)	Neutral	0.18	—
(10,0)	Relatively fair	0.09	1.00

Source: Falk, Fehr, and Fischbacher (in press).

Conclusion

- The Ultimatum data falsify the assumption that players maximize their own money earnings as clearly as experimental data can
- Are the rejections due to the pure self-interest part of the hypothesis, or to the game-theoretic reasoning part or both.
- Camerer argues that it is due to both
 - Deciding to accept an ultimatum offer requires no strategic thinking, so the data reject the own money motivated behavior assumption
 - Direct evidence on how people think discussed in Chapter 4 and 5 show that game-theoretic reasoning is limited.